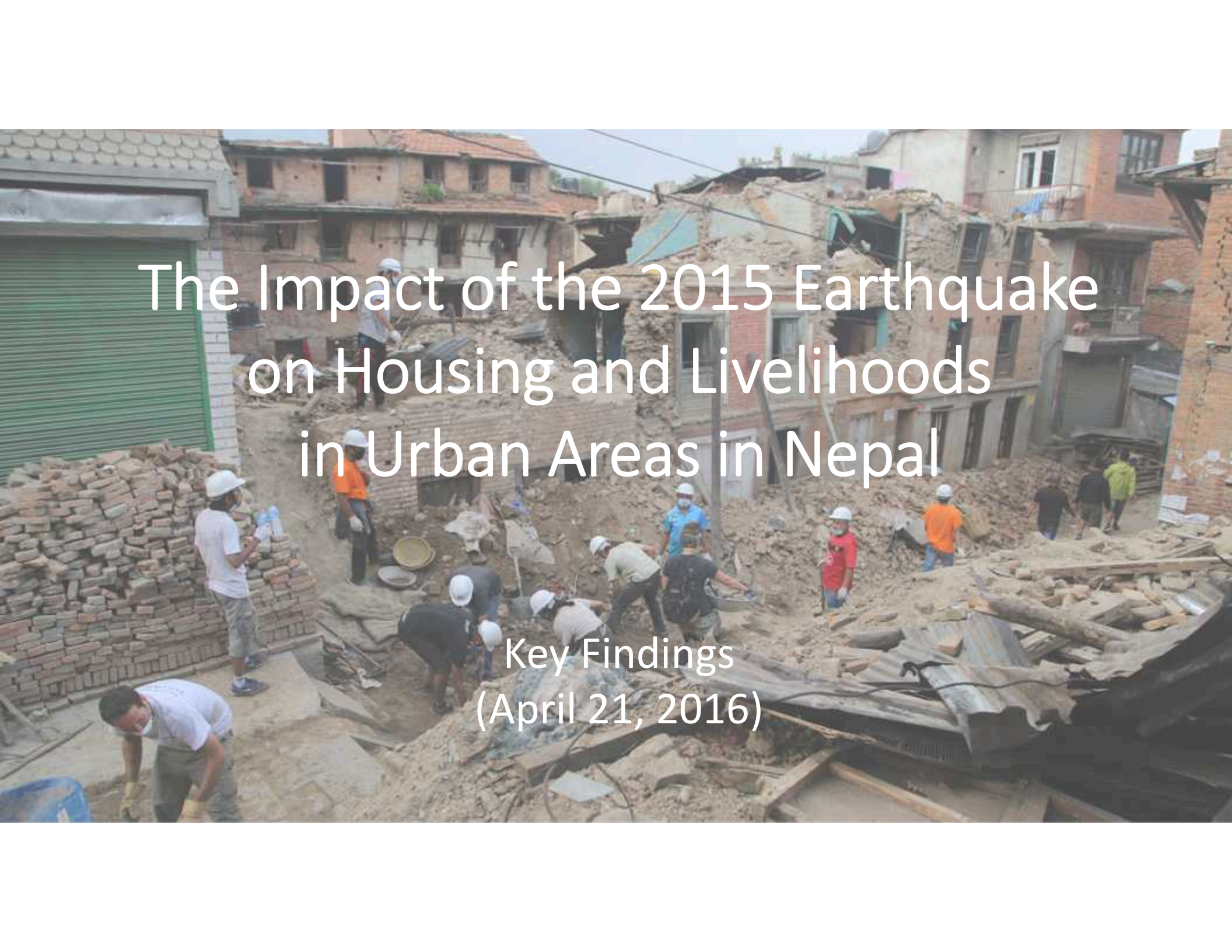




The Impact of the 2015 Earthquake on Housing and Livelihoods in Urban Areas in Nepal

Key Findings
(April 21, 2016)

ISSUES

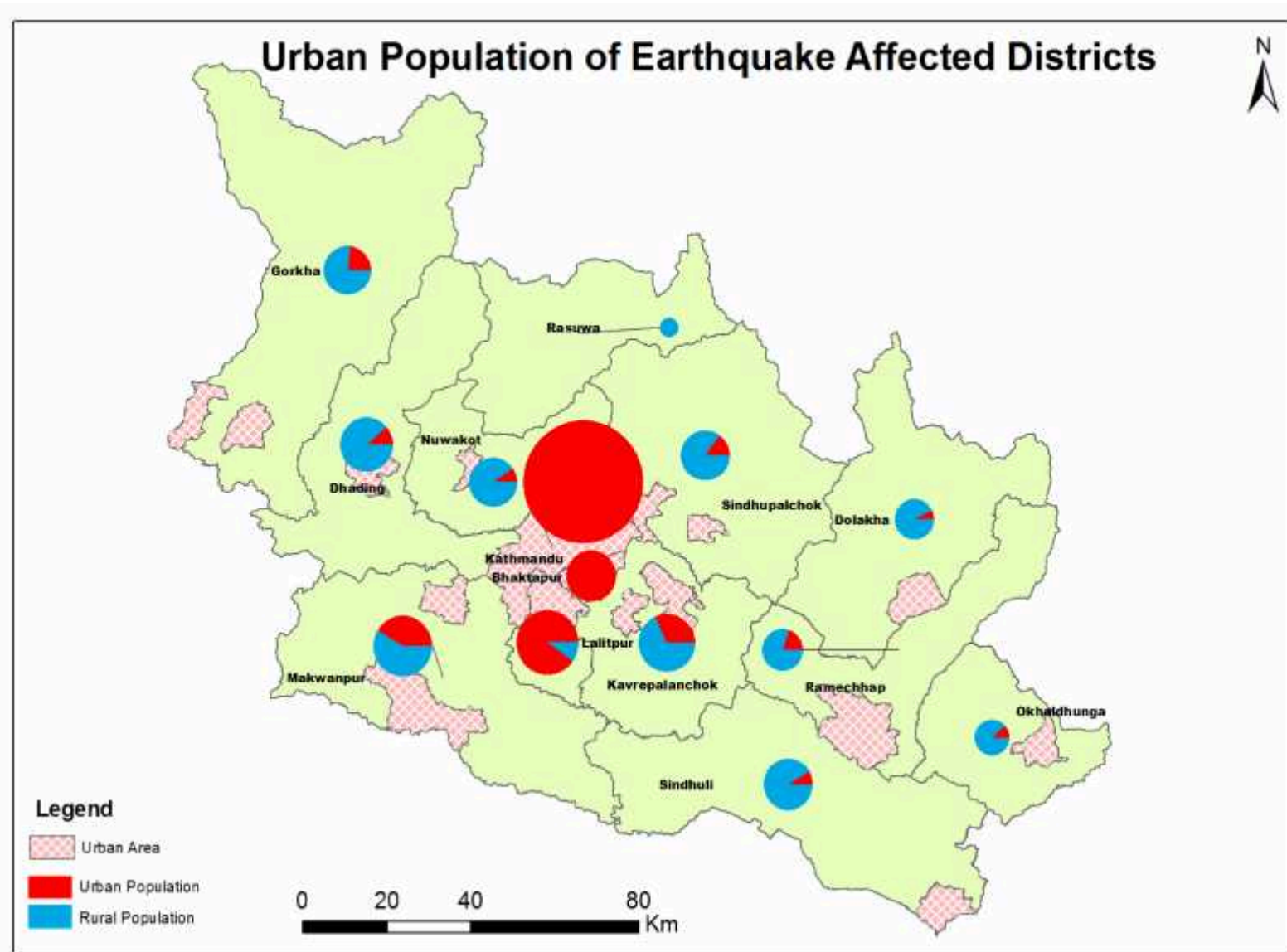
- Methodology
- Impact of Earthquake
- Relief & Recovery Initiations
- Reconstruction Status
- Challenges of Urban Reconstruction
- Financial Features in Urban Reconstruction
- Long Term Impact in Urban Development

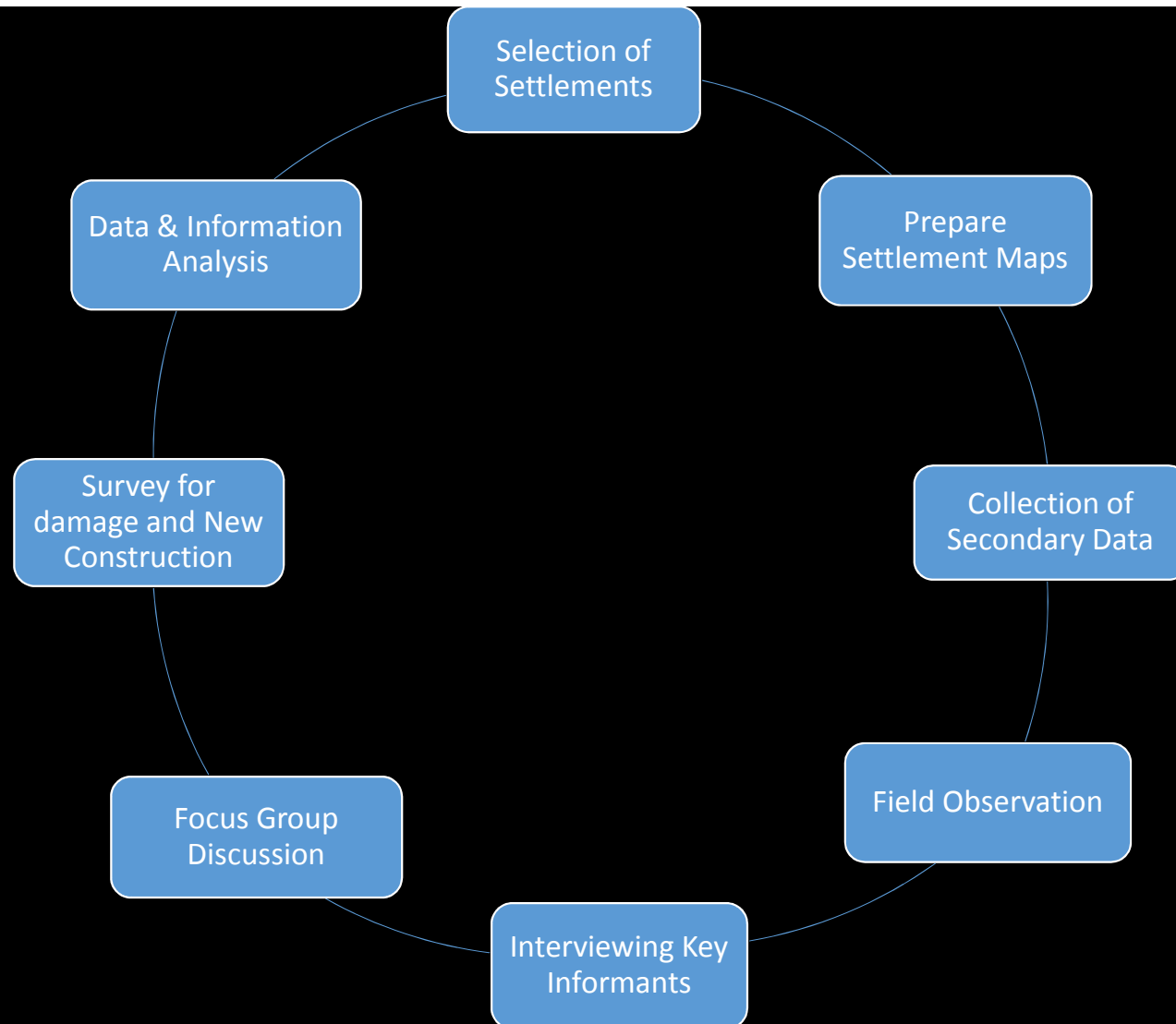
Main Objective

- Analyse the impact of the disaster in the urban sector thereby help to the development of an urban housing recovery and reconstruction policy.

Specific Objectives

- Document the EQ impact on urban communities' housing conditions and livelihoods in selected number of urban centres and their coping mechanism;
- Document the EQ response in the selected urban centres with particular emphasis on those by municipal authorities, urban communities, CBOs and CSOs and support agencies;
- Analyse urban communities' needs and challenges to move forward with recovery and reconstruction with particular emphasis on the vulnerable people.







List of Urban Areas with number of FGDs and Major community

Urban Settlement	Coverage (Ward no.)	Number of FGDs (Total 31)
1. Kathmandu Valley Core Settlement		
Kathmandu - Bhitri sahar	12, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 30	5
Bhaktapur - Bhitri sahar	1 to 17	3
Patan -Mangal Bazar periphery	6, 7, 8, 9, 11, 12, 16, 17, 18, 19, 20, 21, 22	3
Thimi -Balkumari	11, 12, 13, 14	1
2. Peri-urban settlement		
Gongabu	11, 12, 13, 14, 15	1
3. Historical Towns at Fringes of KV		
Bungamati	1, 2, 3, 4, 5, 6	2
Khokana	6, 7, 8, 9	2
Panga	9, 10, 11, 12	2
Shankhu	8, 9, 10, 11	2
4. Market Town out of Kathmandu Valley		
Gorkha Bazar	1, 2, 3, 6	2
Charikot Bazar	1, 10	2
Chautara Bazar	3, 4, 7	2
5. Densely settled area	Ghattekulo	1
6. Informal Settlements	Banshigaht, Khadipakha	2
7. Apartment	Sunrise Apartment	1

Major Focus Community/settlement for FGD

- Heritage conservation area, e.g., Hanuman Dhoka, Mangal Bazar, Bhaktapur Durbar Square
- Heavily damaged clusters of core settlements
- Victims residing in damaged houses
- Poor renters
- Displaced people from the core city
- House Pooling Group (Pilachen & Kilagal)
- Displaced community residing in temporary shelter,
- Disadvantage and marginalized community
- Residents of Informal settlements – Banshighat, Khadipakha
- Residents of damaged Apartment Buildings

KATHMANDU VALLEY CITY CORE

Kathmandu Metropolitan City Core Area



Lalitpur Sub-Metropolitan City Core Area



Bhaktapur Municipality Core Area



Madhyapur Thimi



PERI URBAN AREA

HISTORICAL TOWN

Bungamati Area



Khokana Area



Panga



MARKET TOWN

Chautara



Gorkha



Charikot



Dense Settled Area

Ghattekulo Area



IMPACT OF EARTHQUAKE

BUILDING DAMAGE

Core Area, Historical Town – Majority of buildings damaged are old and made of Mud Mortar

Peri-urban, Market Town – Majority of damage buildings are newly constructed RCC buildings

Densely Settled Area – No Significant damage

Informal Sector – No significant damage

Apartment - Non Structure wall damage

SHELTER

Core Area – Majority of the victims are staying in damaged houses, others are in rented or relatives house

Historical Town – Displaced victims are staying in Temporary Shelter in community rented land but still using basic facilities of damaged houses such as toilet, water and for other privacy issues; remaining are still living in damaged houses

Peri-urban – Staying in rented houses and relatives

Market Town – Living in temporary shelter in their own land

Apartment – Majority are living in Rented house and few are residing in damaged apartment after minor repair work as they do not have other option

LOSS OF INCOME

Core Area – Majority of victims loss / decrease in their income from small scale business, rent and jobs; expenses increase for rent and other basic services

Historical Town – Loss of income due to damage workshop and heavily decreased outside visitors specially in Bungmati and Khokana; loss of small scale business and decrease in agriculture production due to use of land for temporary shelter

Peri-urban, Market Town – Loss of income from house rent and business and increase expenses for renting accommodation

Informal Sector – Loss of services due to displacement of near by residence

Apartment – Increase in expenses for renting house, decrease in value of property

HEALTH, EDUCATION & SAFETY

- Mental anxiety among people due to loss of shelter
- More than 30 people claimed to be dead due to life hardship in Khokana
- Children have been dropped out from the school due to loss of family income in historical town
- Safety and security issue staying in damaged and temporary shelter

CULTURE & HERITAGE

Core Area, Historical Town –

- Several important heritages such as shrines, temples, stone spouts, patti, dyochan, court yards and squares including traditional private houses are damaged;
- Traditional valuable artifacts such as carved doors and windows are either damaged or being sold
- Jatra, festivals and rituals are celebrated just for continuation without enthusiasm, some communities did not celebrate,
- Celebrating important festival such as Indrajatra, Bisket, Rato Machhendranath severely affected due to temporary support (Teko) for damaged buildings
- Regular bhajans are interrupted due to damage of pattis

Peri-urban, market town, informal sector and apartment –

- Rituals and social gatherings are decreased

POOR RENTERS & INFORMAL SETTLERS

- Remained in vulnerable situation where buildings are not yet demolished or moved in vulnerable buildings nearby to secure their existing job and avoiding higher rent
- Many poor renters and informal settlers have loss their job due to displacement of local residence and small shops in neighborhood
- School dropout of children of poor renters due to displacement

RELEIF & RECOVERY INITIATIVES

Core Area, Peri-urban – Few National NGOs, Citizen Forum and local clubs have supported for relief materials and temporary shelter immediately after earthquake, no INGOs are involved

Historical Town & Market Town – NGOs, INGOs, local clubs and citizen forum supported for relief materials including temporary shelter

Almost all affected households have received Government's Support of Rs 15,000 & Rs. 10,000

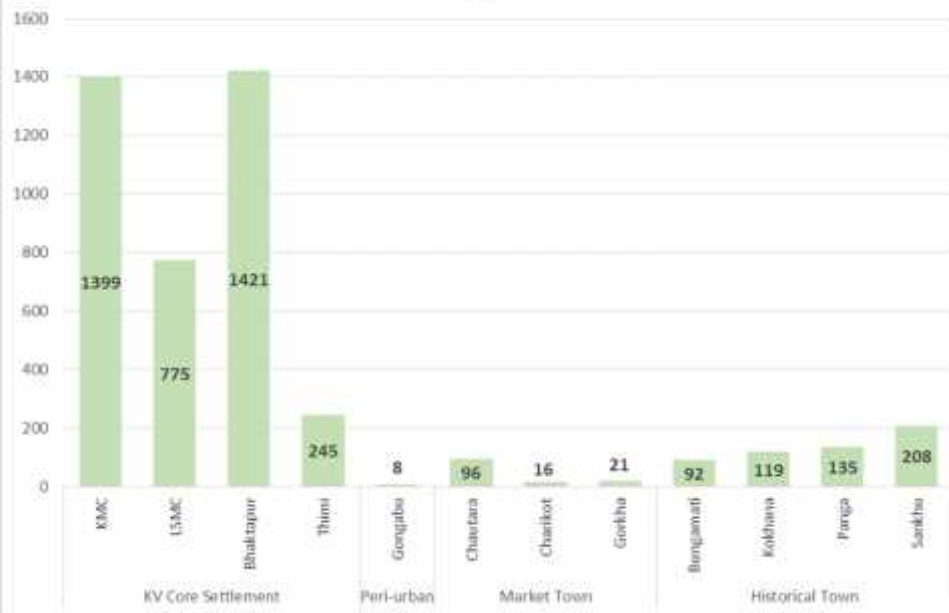
RECONSTRUCTION STATUS

RECONSTRUCTION STATUS

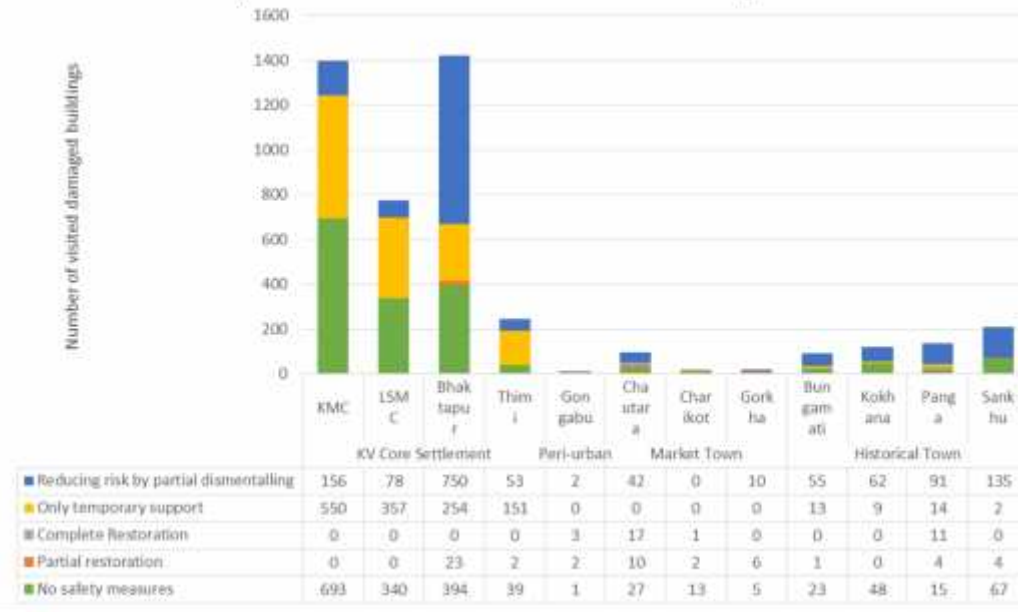
Clearance of site	- Debris of numerous houses is yet to be managed; - Dismantling of vulnerable houses is yet to be carried out
Vulnerable neighborhoods	- Temporary supports (teko) in many buildings thereby disturbing pedestrian and vehicular movement; - Many inhabitants and their business activities are running from vulnerable structures; - Vulnerable spaces are being rented to urban poor
Reconstruction strategy (GON) & Municipal initiation	- PDNA and NRA Act has clearly given indicators for reconstruction; - Municipalities are given responsibility for formulation of rules and regulations on house pooling (?) and urban regeneration (?) but they did minor modification on bylaws only ; - Started issuing building permit (<i>KMC - 94, LSMC – 217, Panga – 22</i>) - Trained masons and provisions of solar panels and rainwater harvesting
Financial support	- NRs. 200,000 (grant) or NRs 1500,000 -2500,000 (soft loan) on capacity basis
Individual/community initiation	- Individual has started new construction but majority are still waiting to see government of Nepal's policy; others could not start due to problems of multiple ownerships and difficulties in dismantling vulnerable structures; - Some communities have come up with different redevelopment plans

UNDEMOLISHED BUILDINGS

Undemolished Buildings Identified in the Field

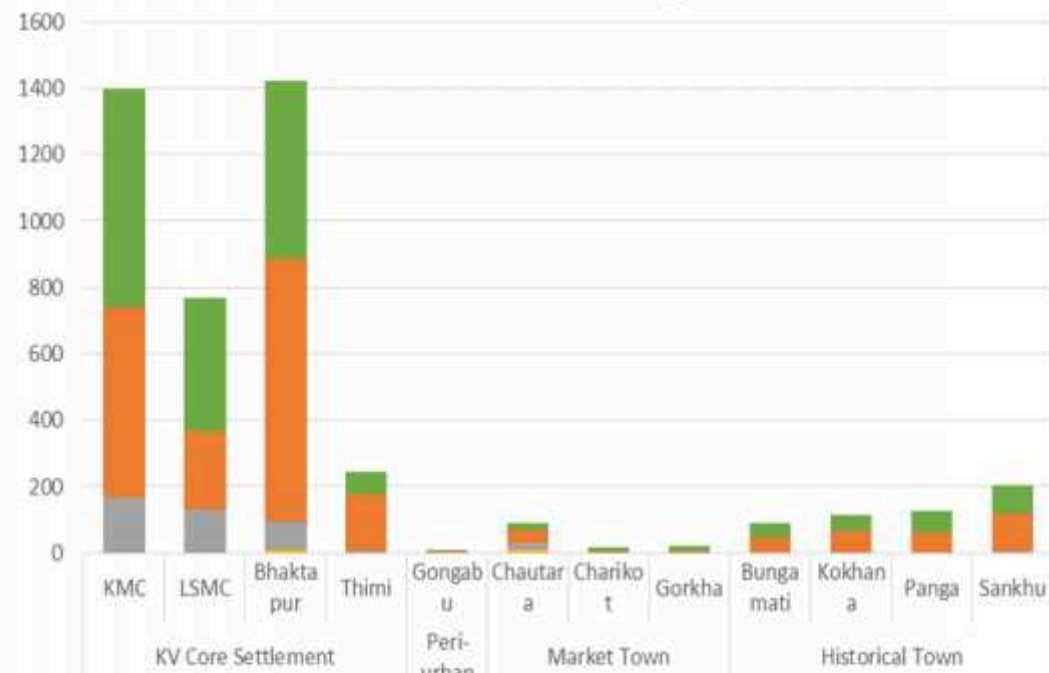


Safety Measures for Undemolished Buildings



Number of visited damaged buildings

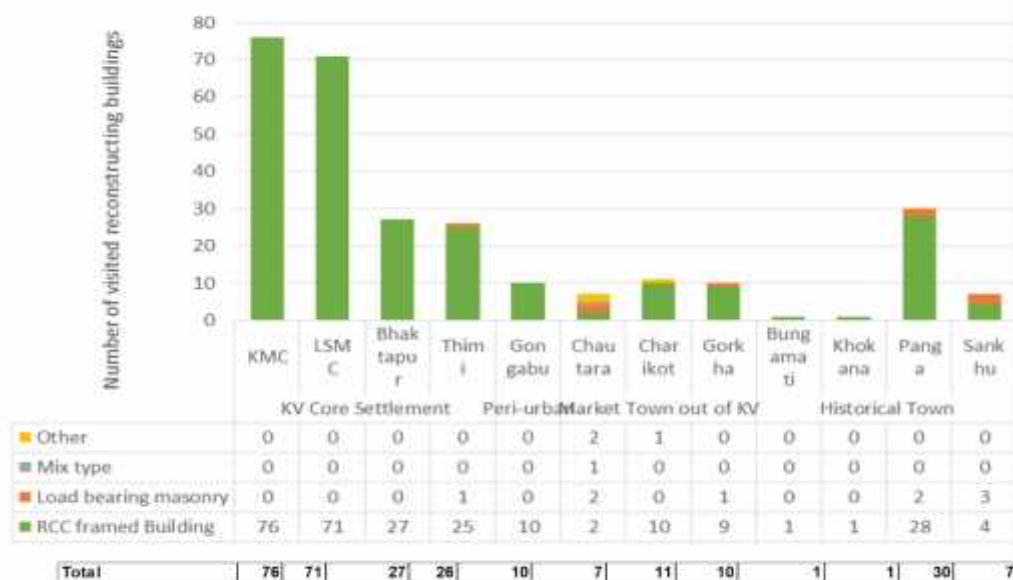
Use of Undemolished Buildings



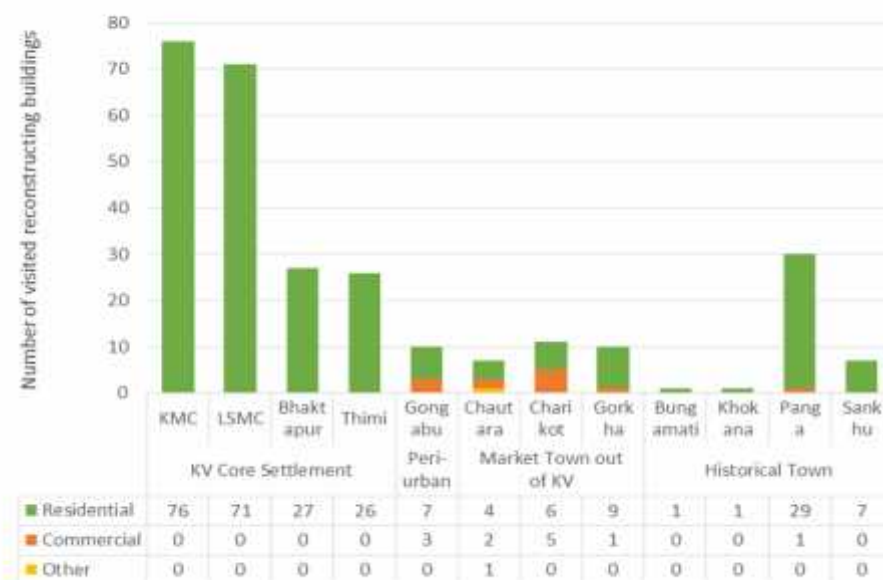
No use	658	401	535	71	2	20	12	14	46	50	67	86
Partial use by Owner itself	572	237	793	169	5	42	3	5	46	66	60	115
Partial use by rentals	169	129	82	5	1	21	1	2	0	0	0	4
Mixed users	0	0	11	0	0	8	0	0	0	0	0	0

NEW CONSTRUCTION

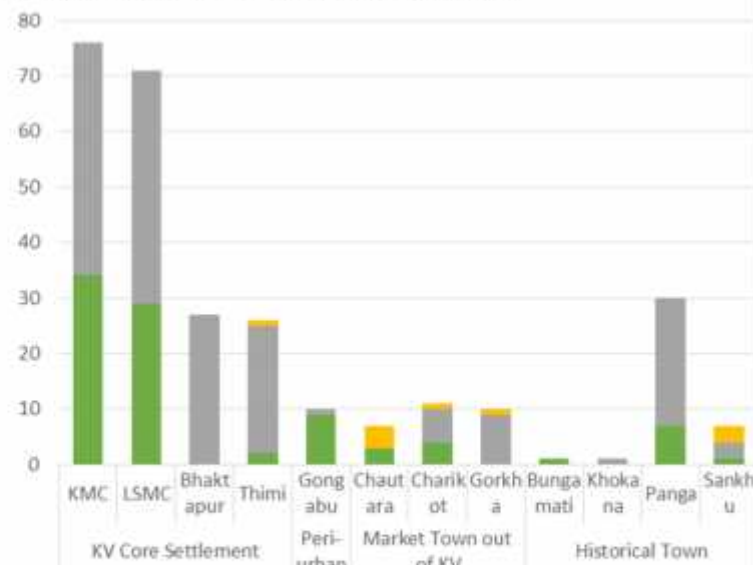
Buildings being reconstructed by Type of Structure



Purpose of building under construction

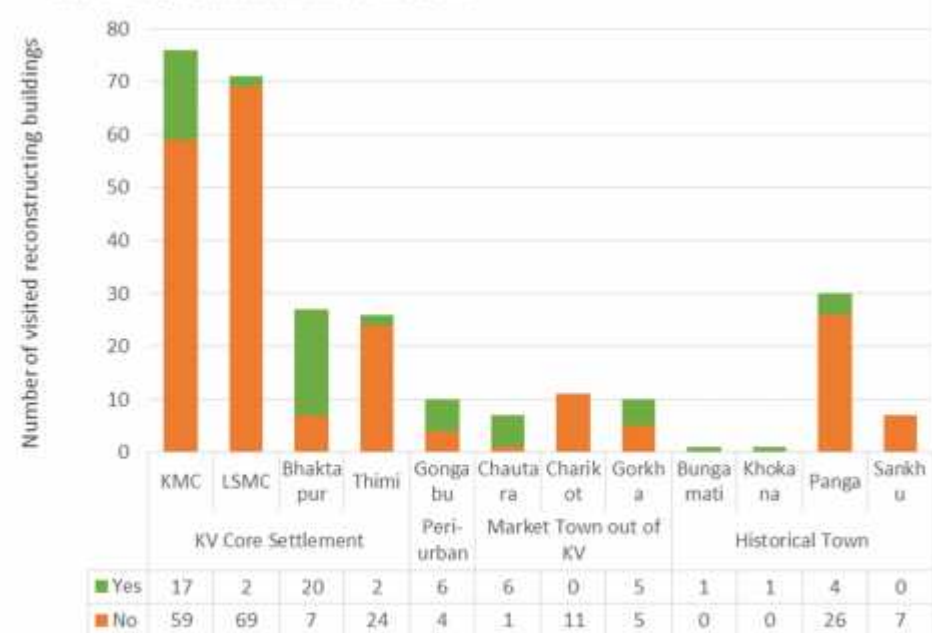


Classification of Building under construction



Low strength masonry Building	0	0	0	1	0	4	1	1	0	0	0	3
MRT limited building	42	42	27	23	1	0	6	9	0	1	23	3
Engineered Building	34	29	0	2	9	3	4	0	1	0	7	1
State of art	0	0	0	0	0	0	0	0	0	0	0	0

Use of EQ Resilience Trained Masons



Yes	17	2	20	2	6	6	0	5	1	1	4	0
No	59	69	7	24	4	1	11	5	0	0	26	7

REASONS FOR NOT CONSTRUCTING

1. Waiting for Government clear policy about reconstruction
2. Lack of Financial Resources
3. Legal issue of land (Specially in case of Khokana, and some areas of KV City Core and other old towns)
4. Risk of dismantling neighbors' buildings (specially in KV City Core)
5. Building permit not issued (Some heritage important areas)

CHALLENGES OF RECONSTRUCTION

Planning and design issues	- What to conserve (community heritage & architectural typologies); - Small and elongated plots without inadequate lighting and ventilation; - Focus on individual buildings without integrated community/neighbourhood planning; - Individual house construction: high cost and unsafe; - Multiple land ownership on plot and house including partition walls; - Issue of building material and construction technology
Permit issues	- No clear rules and regulations for combined housing/house pooling - Confusion over Joint apartment Act and building bylaws (for combined housing) - Linking with GON's financial support is not clear
Financial issues	- Inhabitants in the core are financially poor at least for huge capital investment; - Unaware on financial planning (given by GON and house reconstruction); - Integration of reconstruction with livelihood issues
Safer issues	- How to build safer neighbourhoods from planning (with emergency facilities, integrated infrastructure and open spaces); - In the urban core and historic towns, issue of joint houses to avoid pounding impact
Cost effectiveness	- How to achieve cost effectiveness in individual building construction basis

Community's Perception of Traditional Layout & Planned Settlement

- In Traditional Layout

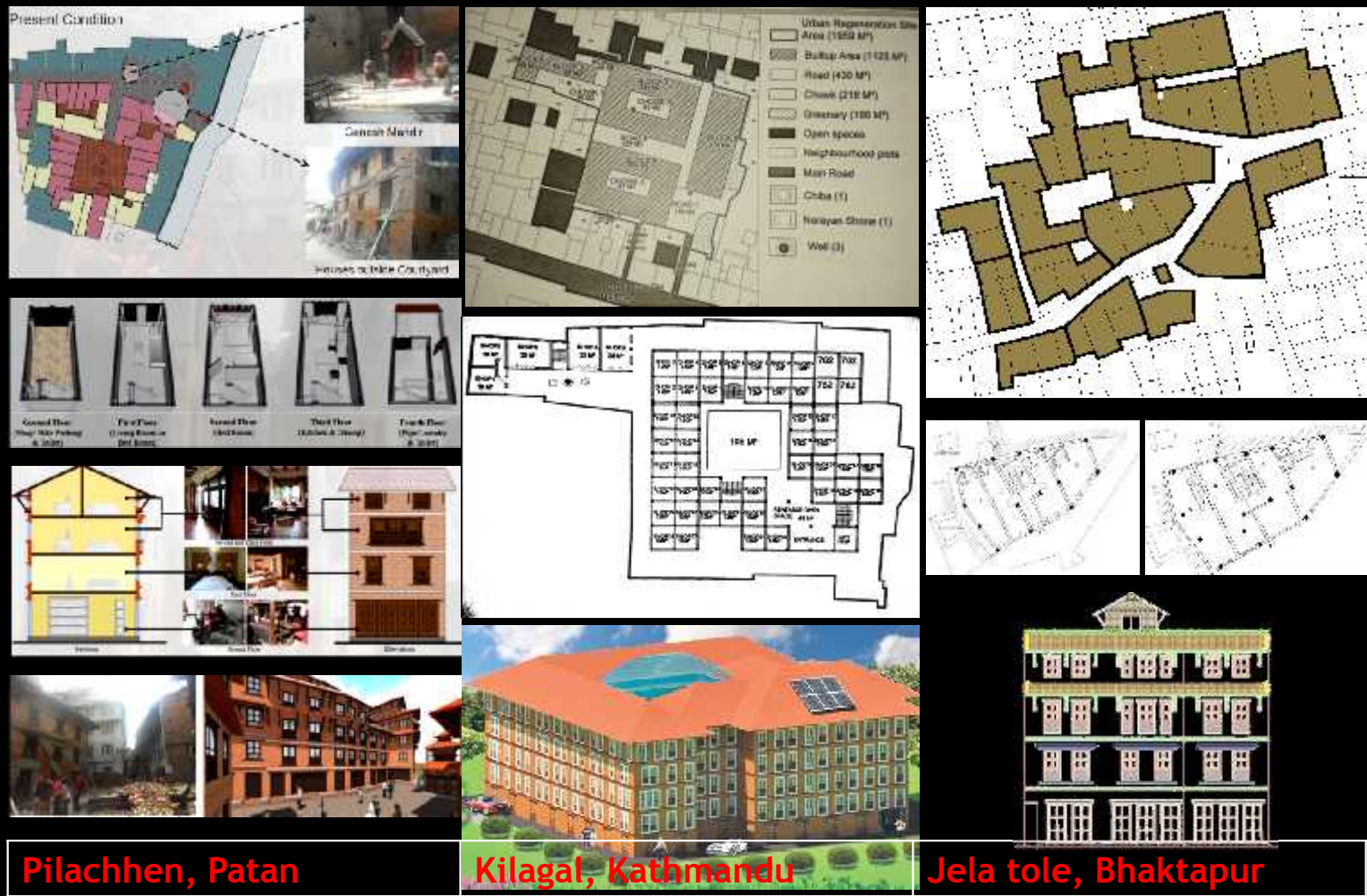
Communities are interested but they cannot afford due to high cost involvement

- Development of Planned Settlement

Interested and few have already started planning and majorities do not know whom to contact and how to proceed

- Plan of restoring private houses in Pilachen, Lalitpur
- Plan of house pooling in Dambo Chowka, Kilagal, Kathmandu
- Local technical people has prepared a Master Plan for development of Khokana and seeking for support
- Bungmati development plan with the technical support of UNHABITAT

Emerging Redevelopment Model



Pilachhen, Patan

Kilagal, Kathmandu

Jela tole, Bhaktapur

Emerging Redevelopment Model - Features

Aspects	Pilachhen, Patan	Kilagal, Kathmandu	Jela, Bhaktapur
Location	Newari core settlement	Newari core settlement	Newari core settlement
House nos.	82	80	66
Cast of community	Maharjan	Maharjan, Gopals & Dongol	Prajapati, Pyatha, Duwal, Suwal, etc.
Professions of community	Agriculture, wood & stone carving, cloth weaving	Agriculture, animal husbandry with some on trade & services	Agriculture with masons, carpenter, painter with few on trade & service
Construction system	Each individual house separately	Single monolithic structure	Combination of small plots into one single unit
Permit system	Existing bylaws	No rules & regulation	Confusion
New use	Lower-guest house & galleries & upper floor for residents (owners)	Lower floors for commercial and upper spaces owners on flat system	Lower floor – galleries, restaurants & home/family stay with upper floors for owners use
Financial system	Owner cash payment = 25% Cash or kind support = 25% Volunteer support = 25% Bank financing = 25%	Rental from commercial uses on lower floors	Combination of owners, business entrepreneurs and heritage conservation including debt financing from BM

FINANCIAL FEATURES

Refinance from NRB for Home Loan

- Banks and Financial Institutions (BFIs) will get refinance facilities at Zero percent interest for a year and renewed thereafter.
- BFIs can not charge more than 2% interest except insurance, collateral valuation, credit information and guarantee fee.
- Refinance will provide minimum 5 years and maximum 10 years with 1 year grace period.
- 2.5 M. & 1.5 M. is maximum single loan size for kathmandu valley and outside valley respectively.
- Only those houses were destroyed in the EQ beyond habitation and none of family members has any other house, are eligible for home loan.
- One BFI can provide maximum 80% of core capital under this scheme.
- Loans will disburse in four installments and must insured.

Financial Grant for House Reconstruction

- To rebuild house will get a relief assistance of NRs. 200,000 plus a concessional loan up to NRs. 300,000.
- Grant will disburse in 3 installments of NRs. 50,000; NRs. 80,000 and NRs. 70,000.
- Those families received aid or new house from NGOs and INGOs can not apply for this grant.
- Family lost two houses are only entitled to received grant for one.
- A single verified victim household is eligible to receive NRs. 200,000 and in special cases NRs. 250,000 even from other NGOs/INGOs

Community Finance

- Majority of victims somehow associated with cooperatives and local informal groups and are getting loans after EQ, and few of them decreased their interest rate.
- Only able to pay interest regularly after EQ.
- Few of cooperatives come up housing loans for their members.
- Lack of capital and cost of fund are major weakness of community finance

Issues related to BFIs concessional Home Loan/ Financial Grant

- Majority of victims are not aware on complete procedure and policy.
- Loans disburse again income of client and collateral, which is difficult as victims lost their income and assets in EQ
- Refinancing facilities is limited only 10 years period whereas other housing loans are available for 15 years as medium term and 25 Years as long term.
- Only who does not have other house than destroyed one is eligible can apply for this scheme made many of destroyed houses are out of this scheme.

Pilachhe Community Housing Reconstruction Initiative

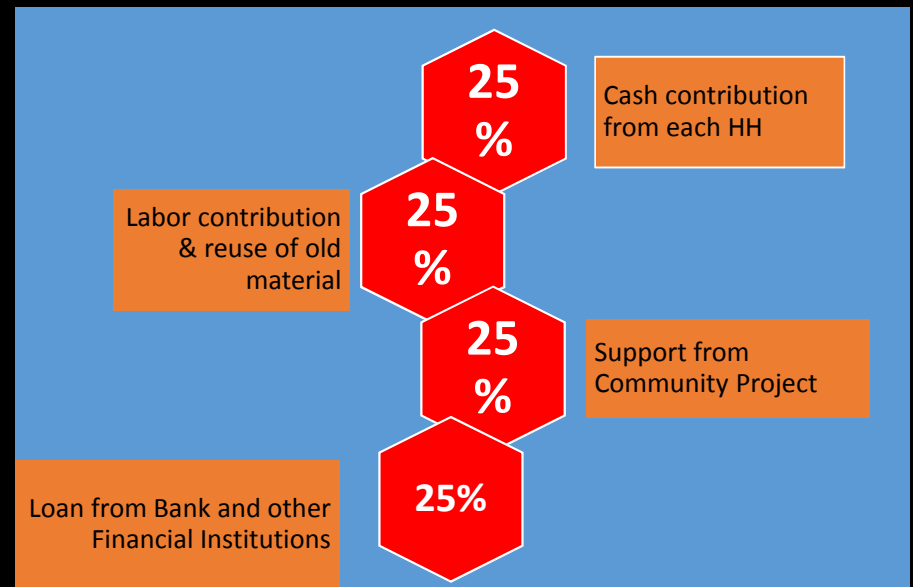
Salient Features

- 82 Households (HHs) out of 110 HHs damages
- Unit Cost: NRs. 3,200 Per Square Feet
- Total Estimated Cost: NRs. 500 M.
- Commitment: around 10% of total cost received
- Set up different Committees: Construction Committee, Procurement Committee, Technical Support Committee, Advisory Committee etc.

Major Challenges of Community:

- Lack of finance as a few HHs are eligible to get concessional support
- Lack of technical personals at the community level

Proposed Financial Structure



Long Term Impact in Urbanization

- Greater chances of converting old and core settlement into slums
- Greater chances of converting temporary shelters into slums
- Chances of adding extra floors for income generation, violating permit is still possible due to weak enforcement of law
- Chances of increasing informal settlers due to displacement of poor renters currently residing in vulnerable buildings

Long Term Impact in Urbanization

- Majority of low income people may get displaced from their original settlement due to un-affordability in constructing new buildings affecting culture and heritage in core and historical towns
- Majority of historical and heritage private houses may vanished because of unaffordable maintenance and reconstruction cost (*800 + only in KMC*) affecting potential tourism industry
- Transformation of Historic Town in concrete structure affecting cultural heritage and tourism potential centers